

Dresden numerical research with PerfectPower

The delivered research object

This release turns declared Dresden-related numerical models into exact executable objects. It supports canonical Long Count conversion, caller-anchored Calendar Round conversion, noncoprime congruence intersections, bounded count and rank selection, source-linked alternative readings, original-day polynomial restrictions, declared Venus correction recurrences, exact long-horizon error envelopes, and complete interval-sum reconstruction inside an explicit finite candidate vocabulary. It reuses PerfectPower's semilinear population counting and polynomial/Sturm domain engine. There is no requirement that the historical integers be perfect powers.

The source modules are `python/perfectpower/dresden.py` and `python/perfectpower/dresden_eclipse.py`. `make dresden` regenerates `receipts/dresden/corpus.json`; `make dresden-test` exercises the new layer. The interface accepts JSON from a file or standard input: `PYTHONPATH=python python -m perfectpower.dresden research/dresden/venus_request.json`. It also imports as an ordinary standard-library Python module. No additional runtime dependency is introduced.

This release combines numerical foundations with complete extraction of selected published tables: 69 eclipse stations, 260 Venus ritual numbers and 8,052 modeled lunar appearances. Published numerical figures have been transcribed; the manuscript itself has not received a complete independent glyph transcription. Candidate-reading and missing-interval demonstrations remain synthetic. Numerical eclipse restart components and interval distributions are implemented, while historical eclipse visibility is not independently validated. Mean-period Venus comparisons are arithmetic experiments rather than ephemeris predictions. The release adds no Lean theorem or end-to-end formal interpreter claim.

Manuscript structure and evidence

Every numerical interpretation needs an evidence record that identifies its source and manuscript location. `solve_alternatives` requires each reading to carry an ID, residue, modulus, source and location. Independent inscriptions each provide a list of alternative readings. The solver considers every combination, intersects its congruences, and retains each compatible assignment with the exact bounded population it describes. Assignments remain separate even if they describe overlapping or identical dates. Their counts must not be added to obtain a distinct-day union.

The Wikipedia overview was examined section by section. Description motivates page/register/column addresses. History motivates separating proposed origin, copying date, European acquisition and interpretation history. Image motivates a manuscript-wide sequence map. Deterioration and pagination motivates preserving witness-specific page order and damaged readings. See also supports comparisons with other codices. References and Bibliography supply source leads; Further reading broadens interpretive context; External links and Video offer facsimile and public-explanation routes. This release records those roles without claiming that bibliographic references have all been read or that images have been exhaustively examined.

A source registry is retained at `research/dresden/sources.json`. It distinguishes the overview from research authorities and from facsimile access. A future transcription must preserve visible numeral marks, inferred numeric values and scholarly interpretations as separate fields. A reconstruction cannot

turn an unreadable image into an observed numeral simply because its preferred arithmetic is attractive.

Long Count as exact integer arithmetic

Digits are most-significant first. The positional weights from the right are 1, 20, 360, 7,200, 144,000 and further factors of 20. Consequently, the second-lowest digit has radix 18 rather than 20.

`long_count_days` rejects noncanonical lower digits and negative values, while permitting an unbounded leading digit. `days_long_count` can use one through 64 places, retaining an unbounded leading place; the two operations are exact inverses on nonnegative integer days within their declared representation.

For example, 4.12.8.0 means $4 \cdot 7,200 + 12 \cdot 360 + 8 \cdot 20 = 33,280$ days. Treating the string as ordinary base 20 would give the wrong result. Leading zeroes may be supplied when canonical; a Gregorian correlation is never inserted implicitly. Signed elapsed differences are ordinary integer differences between nonnegative encoded dates.

Calendar Round and noncoprime compatibility

The default arithmetic epoch convention is 4 Ajaw 8 Kumk'u, encoded by number 4, day-sign index 19 and Haab day 348. Indices run from 0 to 19 for the day signs, with Imix at 0 and Ajaw at 19. Haab is supplied as a linear day index 0 through 364; the month-name layer is not implemented. These conventions are replaceable through the `origin` argument.

A requested number, sign and Haab position becomes three simultaneous congruences modulo 13, 20 and 365 relative to that origin. The first two define a unique class modulo 260. The Haab condition is compatible precisely when it agrees modulo $\gcd(260, 365) = 5$. A compatible Calendar Round therefore identifies a class modulo $\text{lcm}(260, 365) = 18,980$. The test suite roundtrips every one of these 18,980 day positions.

For general progressions $d \equiv a \pmod{m}$ and $d \equiv b \pmod{n}$, let $g = \gcd(m, n)$. A solution exists if and only if g divides $b - a$: necessity follows by subtracting the equations; sufficiency follows by solving $(m/g)k \equiv (b - a)/g \pmod{n/g}$, where m/g is invertible. The returned progression has period mn/g . The implementation handles signed days, modulus one, incompatible phases and arbitrary positive integer moduli without expanding a residue table.

In a closed interval $[L, H]$, the count is $\text{floor}((H - a)/m) - \text{floor}((L - 1 - a)/m)$, with zero returned for reversed endpoints. The first day is $L + (a - L \bmod m)$; rank r selects this day plus rm whenever r lies in the exact count. The corpus counts $10^{30} + 1$ joint-cycle dates over $[0, 37,960 \cdot 10^{30}]$. Storage is independent of that population size.

A direct bridge to polynomial machinery

`polynomial_calendar` accepts an original-day Boolean polynomial sign predicate and a calendar progression $d = a + mk$. It substitutes $a + mk$ into every polynomial before invoking PerfectPower's exact integer polynomial-domain engine. The accepted k intervals are clipped to $\text{ceil}((L - a)/m)$ through $\text{floor}((H - a)/m)$, counted, and transported back to days. `polynomial_calendar_select` recovers the original day for a rank.

This matters when the date constraints are combined with polynomial inequalities from a declared approximate model, uncertainty envelope or bounded research hypothesis. It also avoids constructing a residue truth table of length m . Even a modulus of 10^{30} can be processed by a small parameter

polynomial. Completeness is relative to the supplied polynomial model: a polynomial approximation to an astronomical observable still needs independently justified approximation bounds.

The retained synthetic window $(d-1,000,000)^2 \leq 50,000^2$ intersects the example Calendar Round in exactly five dates: 965,780, 984,760, 1,003,740, 1,022,720 and 1,041,700. It is a declared polynomial window, not a historical uncertainty estimate.

The focused regression compares transported populations against exhaustive original-day evaluation and includes a huge-modulus case. Existing Sturm certificates remain Python replay objects; this application does not add a formal source-to-native proof.

Venus: a sourced arithmetic reproduction

The corpus selectively recomputes correction arithmetic discussed in Gerardo Aldana's 2016 paper, **Discovering Discovery: Chich'en Itza, the Dresden Codex Venus Table and 10th Century Mayan Astronomical Innovation**, especially printed pages 65–69. The registry links the openly accessible paper. Its historical reading is attributed to that author rather than asserted as the sole interpretation.

The canonical arithmetic uses 584 days per round and a 37,960-day Great Cycle. Five rounds equal eight 365-day years: $5 \cdot 584 = 8 \cdot 365 = 2,920$. Sixty-five rounds give 37,960, a multiple of both 260 and 365. The recorded correction-interval values used here are 33,280, 68,900, 185,120 and 9,100. With correction factor 2,340, their integer representations are respectively $(x,y)=(1,2),(2,3),(5,2),(4,61)$ in $x \cdot 37,960 - y \cdot 2,340$. Each identity is recomputed.

The first two corrected spans become 71,240 and 106,860 days after adding one Great Cycle. Their ratios to two and three Great Cycles agree at 137/146. Dividing by their respective 122 and 183 Venus rounds gives the same exact mean, 35,620/61 days. Transporting the supplied base date 9.9.9.16.0 yields 9.19.7.14.0 and 10.4.6.13.0 respectively, reproducing the relevant Long Count arithmetic.

An apparent source typo is retained explicitly in the receipt. For the third correction interval, the printed decimal expression uses $3 \cdot 37,960 - 2 \cdot 2,340$, which evaluates to 109,200 rather than 185,120. The preceding Long Count expression uses coefficient 5, which is consistent with 185,120. The receipt preserves the printed discrepancy and the independently computed consistent coefficient; it does not silently rewrite the source.

Executable correction hypotheses and complete error envelopes

VenusRule declares a block of R rounds taking D days. For round $n=qR+r$, with $0 \leq r < R$, the execution day is $\text{anchor} + qD + 584r$. Repeating this rule indefinitely is an explicit computational experiment; it is not inferred to be the author's complete historical protocol. The uncorrected 584-day rule is labelled synthetic, and the two sourced correction blocks are labelled literature interpretations.

Against a supplied positive rational constant mean P , the elapsed error is $q(D-RP) + r(584-P)$. The default reference is the decimal mean 583.9214, represented exactly as a rational. This measures drift against a mean model, not heliacal visibility, conjunction time or an ephemeris. Correlation constants, observer location, atmosphere and the variable physical synodic interval remain external inputs.

There is no need to enumerate a long run. Error is affine in q and r over every full block. Its extrema occur at the corners of the rectangle of completed blocks, or at the endpoints of the final partial block. `error_envelope` evaluates at most six candidate rounds, including the first and last full-block boundaries and the final round. The corpus executes a horizon of 10^{12} rounds. Tests compare the envelope

against exhaustive execution at horizons around both correction boundaries and for reference means below, equal to and above 584.

Reconstruction and identifiability

`reconstruct_intervals` takes a finite set of nonnegative integer candidates at each labelled slot and an exact target sum. It builds prefix sum populations by dynamic programming. The weight at a sum counts all assignments reaching it. A backward pass retains precisely those values at each slot participating in at least one complete assignment. A witness is returned when one exists; uniqueness means the assignment count is exactly one. Duplicate numeric values in a slot count as one candidate value, not distinct palaeographic readings. Distinct reading identities belong in the alternative-reading layer.

The forward recurrence is $C_{i+1}(s) = \sum_{v \in A_i} C_i(s-v)$, starting at $C_0(0) = 1$. Induction proves that its weights count all assignments. Backward reachability intersects prefix support with the suffix requirement, so a retained value has a full completion and every full completion is retained. Nonnegative choices justify pruning totals above a nonnegative target. A negative target produces an empty population. Explicit work budgets raise an error before any complete result is returned when exceeded.

The proposed standard eclipse span is 11,960 days over 405 lunations, yielding 46 ritual cycles and mean 2,392/81 days. The surviving published table instead ends at 11,959 days; the source audit below keeps that distinction explicit. To demonstrate identifiability, the corpus separately declares a synthetic vocabulary of 69 labelled slots, each allowed 148, 177 or 178 days, summing to 11,960. This admits exactly 3,048,179,358,545,703,294,215,589,000 assignments. It establishes that a total and interval vocabulary alone cannot determine an ordering. It does not establish that any of those assignments is a historical table.

External resources before implementation

The source-first expansion used Aldana's full 2016 paper, the full Justeson–Lowry 2025 article in JATS XML, its numerical figures and supplementary methods, both original S1/S2 workbooks, the complete 2020 Venus tables, and Förstemann facsimile packets covering pages 13–24 and 46–59. Facsimile pages 24 and 46 were visually inspected. Acquisition receipts retain source URLs, signatures, byte counts and SHA-256 hashes. Bulk facsimile packets are not redistributed. `scripts/acquire_dresden_sources.py` can acquire the resources again; its network timeout is configurable. `scripts/ingest_dresden_lunar.py` reads cached numeric XML directly from the original XLSX files without Excel, third-party runtime packages or edits to the workbooks.

The 2025 article and supplementary resources are attributed to John Justeson and Justin Lowry, DOI <https://doi.org/10.1126/sciadv.adt9039>. The article's Creative Commons Attribution 4.0 license covers the retained article XML, figure copies and supplement files. Full attribution and source-specific licensing appear in `research/dresden/NOTICE.md`. Aldana's full paper is linked rather than redistributed. Juárez Nájera and Castellanos's 2020 paper is linked at <https://doi.org/10.19130/iifl.ecm.2020.56.2.0004>; only its numerical labels are extracted into the working table.

Complete published numerical table audits

Figure 1 of Justeson–Lowry supplies all 69 station records, each with cumulative lunation index, cumulative day value and the authors' intended/contrived classification. The extracted table has 55 intended and 14 contrived stations, terminates at month 405 and day 11,959, and contains derived increments 148 days nine times, 177 days 53 times and 178 days seven times. Fourteen selected raw cumulative numeral readings and their stated corrections are retained and independently checked. Source coordinates use the published station IDs; manuscript page boundaries are not guessed.

The source figure reports 147 days for station 19, whereas its consecutive cumulative values give $3,278 - 3,130 = 148$. Both the printed increment and the derived increment remain in the source record, and the audit reports the difference. Two proposed standard-table alternatives insert one additional day beginning at surviving day 7,619 or 7,796, producing first shifted values 7,620 or 7,797 and terminal day 11,960. These remain proposed alternatives, separately labelled from the surviving table.

The 2020 publication's Tables 1 and 2 give 260 ritual-number labels across pages 46–50, thirteen rows and four columns. All labels are extracted with publication coordinates and compared against a declared 1 Ajaw entry, the phase intervals 236, 90, 250 and 8 days, and a 584-day round. Exactly 259 labels match this canonical numerical model. Page 48, row 1, column D reports 1 where the model gives 11. This is a publication/model disagreement, not an independent determination of the manuscript glyph. The supplied label is preserved.

The full supplied lunar interval census

Table S2 supplies 8,052 modeled lunar crescent appearances. The supplementary methods describe a visibility model using astronomical calculations and observer assumptions. These appearances are model outputs, not ancient recorded observations, and this release does not independently rerun their astronomical or visibility calculation. The importer retains cached decimal JDN strings and uses $\text{floor}(\text{JDN})$ to obtain strictly increasing integer dates. Table S1's original category cells and cached frequencies are likewise retained, including duplicate categories.

For every span n from 1 through 405, the census counts every overlapping pair difference $d_{(i+n)} - d_i$. This counts 3,178,845 intervals in total. Histograms and all frequencies are exact integers and rational numbers. The modal proportions for 49, 81, 98 and 405 months are respectively $3,768/8,003$, $5,333/7,971$, $7,081/7,954$ and $6,242/7,647$, reproducing the corresponding headline frequencies. For 405 months, the full distribution is 11,959 days in 1,117 cases, 11,960 in 6,242 cases and 11,961 in 288 cases.

The comparison checks each of the 405 source rows with a tolerance of 10^{-12} for the spreadsheet's cached decimal frequencies. Exactly 387 rows agree. The disagreeing spans are 60, 90, 91, 96, 113, 154, 183, 196, 254, 273, 282, 305, 306, 335, 341, 388, 391 and 392 months. The receipt records duplicate categories, omitted categories and individual frequency differences separately. Both reported and recomputed values remain available. This is agreement under the stated extraction convention; it is not a blanket rejection of the paper's historical conclusions. The complete census and comparison are in `receipts/dresden/lunar_census.json`.

Exact overlapping restarts and bounded drift policy

The restart engine places the 55 intended station months at every base generated by a declared repeating restart pattern. For four 358-month steps and one 223-month step, the base pattern has period 1,655 months. Overlapping table predictions are deduplicated by residue, while each residue's earliest activation is retained. This gives 227 eventual prediction residues and density $227/1,655$. The startup boundary remains exact: a residue contributes only after its first occurrence. Existing semilinear counting and selection operate on this domain, including ranks of 10^{30} . These are predicted lunation indices, not a count of physically visible eclipses.

Figure 8 supplies rounded mean offsets of +0.0972 days per 358-month restart and -0.42356 days per 223-month restart. The terminal offset for their five-step combination is -0.03476 days regardless of ordering. Under precisely these supplied offsets, a bounded minimax dynamic program checks all orderings and returns the unique sequence 358, 358, 223, 358, 358. Its maximum absolute prefix offset is exactly $5,729/25,000$, or 0.22916 days. This is a mathematical optimization under rounded mean offsets; it does not establish that an ancient practitioner selected that order.

The first dynamic-programming pass minimizes the maximum prefix magnitude at each lattice state. A second pass counts every path staying within the final global bound; retaining only prefix-minimal predecessors could incorrectly discard paths that still attain that final bound. Enumeration tests check both the optimum and the number of attaining paths. The separate mean-period drift function uses caller-supplied rational lunar and nodal means; its outputs are not silently identified with the differently rounded figure offsets.

An interactive research instrument in React and Three.js

The companion Dresden Codex Observatory turns the numerical corpus into six linked visual views. The calendar instrument advances elapsed days through a 37,960-day joint cycle, with selectable exact day inputs and animated rings for the 13-number, 20-sign, 365-day Haab and 584-day Venus phases. Its epoch-aware date readout remains separate from the rings' elapsed angular phase. The display is coordinate geometry, not a claim that this instrument existed historically. Calendar Round population queries and Long Count conversion use JavaScript BigInt, including examples at 10^{30} days and tests at 10^{100} .

The eclipse view maps all 69 published stations onto a 405-month circular coordinate, preserves the intended/contrived classification, highlights the station-19 increment disagreement and permits selection of both proposed standard-table alternatives. The Venus view exposes every one of the 260 supplied labels at its publication coordinates. A separate chart illustrates the three declared correction recurrences against the constant reference mean. The lunar view makes all 405 interval distributions selectable, displays exact histogram counts and fractions, and identifies all eighteen published spreadsheet disagreements. Source and recomputed values remain separately visible.

The restart view compares all five possible positions of the short restart, including their shared terminal offset and distinct intermediate drift. Its overlap chart displays the 55 intended stations at each declared base. A browser population query retains the first activation of every residue, deduplicates repeated predictions and counts exactly at arbitrary input horizons of up to 120 decimal digits. The eventual residue count is recomputed for the selected ordering rather than copied from a different policy. The sources view retains the manuscript section map, primary references, published figures and historical facsimile images.

The browser source is in `web/dresden-observatory`. Three.js WebGPURenderer prefers WebGPU and automatically falls back to WebGL 2 when supported. The React interface remains available if GPU

initialization fails. Its scene supports camera rotation, zoom, reset and clickable eclipse stations, with equivalent keyboard-accessible station buttons. The calendar animation starts only on request. Responsive layouts preserve the numerical controls and source inspections on narrow screens. Optional WebMCP registration exposes the same view/day/station/span actions only in browsers supporting that proposed API.

The production React build and a separate static build pass. The self-contained HTML companion embeds its bundled React/Three.js code, styles, numerical datasets, attributed source images and PDF monograph, avoiding runtime CDN dependencies. External research links remain external. The HTML's embedded module passes JavaScript syntax checks and has no external script, stylesheet or module-preload dependency. The browser arithmetic is compared with independently generated Python fixtures for 25 day values up to 10^{100} , all five overlap orders at seven horizons, exhaustive small noncoprime congruences, enormous population counts and malformed inputs. The result is retained in receipts/dresden/web-math-parity.log. Browser/GPU execution and optional WebMCP validation were unavailable in the build environment; no rendering-performance benchmark is claimed.

A second research layer across all six views

The expanded calendar view solves user-selected ritual number, day sign and Haab congruences, optionally adding any of the 584 Venus phases. Inclusive bounds and zero-based rank are explicit inputs. The response distinguishes incompatible congruences, a compatible progression with no bounded members, and a rank outside a nonempty population. It reports residue, period, first/last matching day, exact count and a selected day. The selected result can be shown on the shared calendar rings. BigInt arithmetic retains exactness, and JSON exports encode large integers as decimal strings rather than unsafe JavaScript numeric literals.

The expanded eclipse view compares every surviving increment with both proposed standard-table alternatives, showing exactly which transition receives +1 and how that change propagates through later cumulative totals. Its complete station ledger filters the authors' intended/contrived classes and published increment discrepancies. A selected raw numeral record is converted with the Long Count weights before its separately stated correction is applied. Fourteen retained raw correction identities are checked; other stations explicitly report that a raw reading was not captured. A station can be inspected as an elapsed day in the calendar view, without inventing an absolute historical epoch for that station.

The expanded Venus view accepts a round horizon, one of three declared recurrences and a positive rational reference mean. Exact execution, terminal error, completed blocks and remainder are displayed alongside the complete error envelope and its attaining candidate rounds. Ninety-six independent Python/browser comparisons exercise all three models at eight horizons and four means, including a 10^{30} -round horizon and means 583.9214, 584, 585 and 35,620/61. At most six boundary candidates suffice regardless of horizon. The correction ledger preserves all four sourced intervals, coefficients, mixed-radix representations, the CI3 printed discrepancy, the common corrected mean and the exact transported anchors.

The expanded lunar view loads the complete original supplied integer-day sequence and cached JDN strings. It recomputes the selected span over any valid contiguous appearance window, returning exact histograms, fractions, mean, population variance, median and discrete 2.5/97.5 percentiles. The central interval is descriptive, not a confidence interval or new astronomical inference. The sample count for a window $[a,b)$ and span n is $b-a-n$. Splitting the 8,052 appearances into two 4,026-appearance halves at span 405 gives 3,621 intervals in each half, whereas the complete corpus has 7,647: the missing 405 pairs cross the split. Thus a naive sum of window counts would not reconstruct the full distribution.

For 405 months, the first half contains 512, 2,978 and 131 intervals of 11,959, 11,960 and 11,961 days; the second half contains 549, 2,927 and 145. Full-corpus mean and variance are exactly 91,457,291/7,647 and 10,056,794/58,476,609 days squared. The interface can rank modal concentrations using exact fraction comparisons and filter all eighteen source disagreements. It also compares centered residual distributions using exact total variation. Centering uses $\text{floor}(n \times 29.530589)$, an explicitly declared reference convention. The 98-versus-405 month distance is 2,759,969/30,412,119; this descriptive distribution difference does not establish historical eclipse performance.

The expanded restart view inverts the distinct overlapping prediction population. A zero-based rank and inclusive starting month select an exact lunation index through monotone integer count queries and binary search, including finite startup and deduplication. It accepts enormous ranks without enumerating the preceding predictions. A separate sensitivity experiment replaces the two rounded offsets with arbitrary rational values and evaluates every order of four long and one short restart. All tied global minimax optima are preserved; the selected order updates the shared overlap schedule. Five Python/browser policy comparisons include both the published defaults and deliberately tied zero-offset inputs.

The expanded evidence view supplies a searchable, typed transformation ledger connecting overview, publication interpretation, numerical extraction, figure transcription, modeled appearances, recomputation and declared experiments. Each record has its source, location, transformation, result and explicitly identified source/code/dataset digest when retained. Exports include the evidence ledger and complete original lunar input extraction. Numerical records, scholarly corrections, alternative reconstructions and physical visibility models remain separate evidence categories. Every feature adds a structured experiment export with exact integer/rational representation.

Independent parity checks cover four calendar queries, 96 Venus executions/envelopes, twenty recomputed corpus windows, four distribution comparisons, twenty ranked predictions and five rational restart policies, plus all fourteen raw corrections. All six React views and their deeper panels are rendered through the React server renderer, checked for their expected content and invalid numeric output. TypeScript, the production build, static build and embedded JavaScript syntax checks pass. A two-dimensional phase projection retains functional geometry and clickable eclipse stations when GPU initialization is unavailable. Browser/GPU execution and optional WebMCP validation remain untested; no new browser performance claim is made. Core Python operations `lunar_window` and `lunar_compare` reproduce the new corpus analyses from checkout data. `make dresden-deep` regenerates the independent receipts.

Verification and next mathematical work

Twenty-four focused tests cover Long Count representations, exhaustive noncoprime CRT intersections, all 18,980 Calendar Round phases, huge populations, source-linked alternatives, polynomial transport, Venus envelopes, reconstruction against exhaustive enumeration, all 69 station records, both proposed standard tables, overlap deduplication and startup, huge rank selection, minimax optimization with all ties, all 260 Venus numbers and the full supplied lunar corpus. Malformed phases, noncanonical digits and exhausted budgets are rejected. The focused log is `receipts/dresden/focused-tests.log`.

The broader repository checks passed after installing the optional `z3-solver` and `python-flint` dependencies: 1,332 core tests with four skips, 14 continuation tests, 10 expert tests and eight Galois tests. This broader run preceded the final source-data additions; the final 24-test focused run verifies those additions. Its complete output is retained in `receipts/dresden/regression.log`. Existing skipped tests remain explicitly recorded. No new Lean proof is supplied or claimed.